Homework 4

ASEN 5335 Aerospace Environment

Justin Le
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Problem 1: Field Magnitudes

Using the WMM model from the NOAA calculator, we find the magnetic field in Boulder, CO to be,

Model Used:	WMM-2025						
Latitude:	40° 0' 59" N						
Longitude:	105° 17' 10" W						
Elevation:	5430.0 ft Mean Sea Level						
Date	**Declination (+ E | - W)**	**Inclination (+ D | - U)**	**Horizontal Intensity**	**North Comp (+ N | - S)**	**East Comp (+ E | - W)**	**Vertical Comp (+ D | - U)**	**Total Field**
2026-10-18	7° 40' 8"	66° 0' 55"	20,760.4 nT	20,574.7 nT	2,770.4 nT	46,662.2 nT	51,072.1 nT
Change/year	-0° 4' 50"/yr	-0° 2' 59"/yr	-8.9 nT/yr	-5.0 nT/yr	-30.2 nT/yr	-129.4 nT/yr	-121.9 nT/yr
Uncertainty	0° 22'	0° 12'	133 nT	137 nT	89 nT	141 nT	138 nT

Figure 1: Boulder Magnetic Field

We repeat with the same model for McMurdo Station Anarctica to find,

Model Used:	WMM-2025						
Latitude:	77° 50' 46" S						
Longitude:	166° 40' 5" E						
Elevation:	33.0 ft Mean Sea Level						
Date	**Declination (+ E | - W)**	**Inclination (+ D | - U)**	**Horizontal Intensity**	**North Comp (+ N | - S)**	**East Comp (+ E | - W)**	**Vertical Comp (+ D | - U)**	**Total Field**
2026-10-18	140° 5' 50"	-80° 5' 50"	10,647.5 nT	-8,168.0 nT	6,830.2 nT	-60,990.6 nT	61,913.0 nT
Change/year	-0° 12' 10"/yr	0° 2' 29"/yr	33.8 nT/yr	-1.7 nT/yr	50.6 nT/yr	66.8 nT/yr	-60.0 nT/yr
Uncertainty	0° 34'	0° 12'	133 nT	137 nT	89 nT	141 nT	138 nT

Figure 2: McMurdo Magnetic Field

TODO: Based on the inclination observed at McMurdo, this tells us that the magnetic fields will pass through the station based on the inclination and the latitude of the station.

Problem 2: Dipole Field

(i).

Assuming a dipole field model with $L = 4$ $B_0 = 31,000$ nT, $R_E = 6371$ km, and at the equator ($\lambda = 0°$), we can use the following equations to calculate the magnetic field magnitude,

$$r = LR_E \cos^2 \lambda$$
$$|B| = B_0(\frac{R_E}{r})^3\sqrt{1 + 3\sin^2 \lambda}$$

$$\therefore r = 25,484,000$$
$$\boxed{\therefore |B| = 484.4 \text{ nT}}$$

(ii).

Repeating the above equations only for $\lambda = 30°$,

$$\therefore r = 19,113,000$$
$$\boxed{\therefore |B| = 1518.9 \text{ nT}}$$

Problem 3: Magnetopause

(i).

To find the standoff distance of the Earth to the magnetosphere, we can use the equation,

$$\rho_{sw} v_{sw}^2 = \frac{(2B_{mag})^2}{2\mu_0} = \frac{4B_0^2}{2\mu_0 L^6}$$
$$B_{mag} = B_0 \left(\frac{R_E}{r}\right)^3 = \frac{B_0}{L^3}, \qquad B_0 \approx 31{,}000 \text{ nT}$$
$$L = \left(\frac{2B_0^2}{\mu_0 \rho_{sw} v_{sw}^2}\right)^{1/6}$$

Given $v_{sw} = 400$ km/s, and $\mu_0 = 4\pi \times 10^{-7}$, and $\rho_{sw} = n m_p = 5$ proton/cm^3 $*1.672 \times 10^{-27}$ kg $= 8.36 \times 10^{-21}$ kg/m^3.

$$\boxed{L = 10.2}$$

(ii).

We can repeat the calculation above but instead using $v = 1000$ km/s, and $n = 10$ proton/cm^3, for a new $\rho_{sw} = 1.67 \times 10^{-20}$ kg/m^3.

$$\boxed{L = 6.7}$$

Problem 4: The Ring Current

(i).

The reason why there is a current in the first place is because of protons drifting west, while electrons drift east. This causes a net westward current which is the ring current. Additionally, there is a gradient force present...TODO FINISH

(ii).

Finding how long it takes for an 85 keV electron to drift all the around Earth at $L = 4$, we can use the equations,

$$\mathbf{v}_{d\nabla} = \frac{mv_\perp^2}{2q}\frac{\mathbf{B}\times\nabla B}{B^3}$$
$$E = \frac{1}{2}mv_\perp^2$$

If we assume the particle remains in the equator and all its energy is perpendicular ($v_\parallel = 0$). At the equator $|\mathbf{B}\times\nabla B| = B|\nabla B|$. The drift speed is then,

$$v_d = \frac{2E}{2q}\frac{B|\nabla B|}{B^3} = \frac{E}{qB}\frac{|\nabla B|}{B}$$

To find the gradient we use the dipole field at the equator ($\lambda = 0°$) from Problem 2 and take the derivative with respect to r,

$$B = B_0\left(\frac{R_E}{r}\right)^3$$
$$\frac{dB}{dr} = -3B_0\frac{R_E^3}{r^4} = -\frac{3B}{r}$$
$$\therefore |\nabla B| = \frac{3B}{r}$$

Substituting the gradient back into the drift speed,

$$v_d = \frac{E}{qB}\frac{3B}{rB} = \frac{3E}{qBr}, \qquad B = \frac{B_0}{L^3}, \quad r = LR_E$$

The time to drift all the way around the Earth is the circumference of the drift path divided by the drift speed,

$$T = \frac{2\pi r}{v_d} = \frac{2\pi qBr^2}{3E} = \frac{2\pi qB_0R_E^2}{3EL}$$

Given $L = 4$, $B_0 = 31,000$ nT, $R_E = 6371$ km, and $E = 85$ keV, we have $B = 484.4$ nT (from

Problem 2) and $r = 25,484,000$ m.

$$v_d = \frac{3(85,000)}{(484.4 \times 10^{-9})(25,484,000)} = 20,660 \text{ m/s}$$

$$T = \frac{2\pi(25,484,000)}{20,660} = 7,750 \text{ s}$$

$$\boxed{\therefore T \approx 2.15 \text{ hours}}$$

Problem 5: Drift due to Gravity

(i.)

Under the influence of gravity, protons will drift east and electrons will drift west.

(ii.)

We find the magnitude of the drift velocity due to gravity is with the equation,

$$\mathbf{v}_{dg} = \frac{m\mathbf{g} \times \mathbf{B}}{qB^2}$$

(iii.)

To find how long it will take to drift around the Earth under gravity, we can use the drift to to gravity equation and assuming a particle drifting around the equator,

$$\mathbf{v}_{dg} = \frac{m\mathbf{g} \times \mathbf{B}}{qB^2} \quad , \quad v_{dg} = \frac{mg}{qB}$$

$$g = \frac{g_{Earth}}{L^2} = \frac{9.81}{4^2} = 0.613 \text{ m/s}^2$$

$$T = \frac{2\pi r}{v_{dg}} = \frac{2\pi qBr}{mg}$$

$$v_{dg} = \frac{(1.673 \times 10^{-27})(0.613)}{(1.602 \times 10^{-19})(484.4 \times 10^{-9})} = 1.32 \times 10^{-2} \text{ m/s}$$

$$T = \frac{2\pi(25,484,000)}{1.32 \times 10^{-2}} = 1.21 \times 10^{10} \text{ s}$$

$$\boxed{\therefore T \approx 384 \text{ years}}$$

Here we find that the drift to to gravity is far slower than the gradient drift from Problem 4, and that when calculating ring current this can be assumed to be a negligible effect.

Problem 6: Geomagnetic Indices

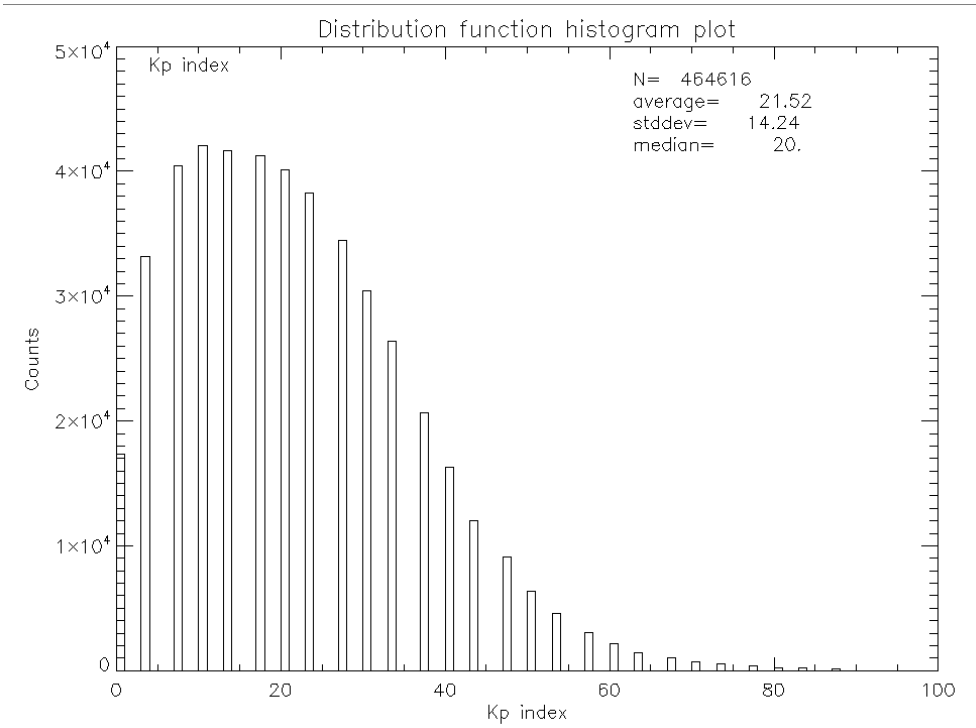


Figure 3: Distribution Function

(i.)

Using Figure 3, we can identify the most common Kp in this distribution as $Kp = 1$, as the most frequent occurrence in this distribution is at 10.

(ii.)

Examining what values will be at level 9, we find the ratio of the Kp = 90 out of the total measurements.

$$\boxed{\frac{39}{464{,}616} = 0.0084\%}$$

(iii.)

When repeating the above for July 2004, we find,

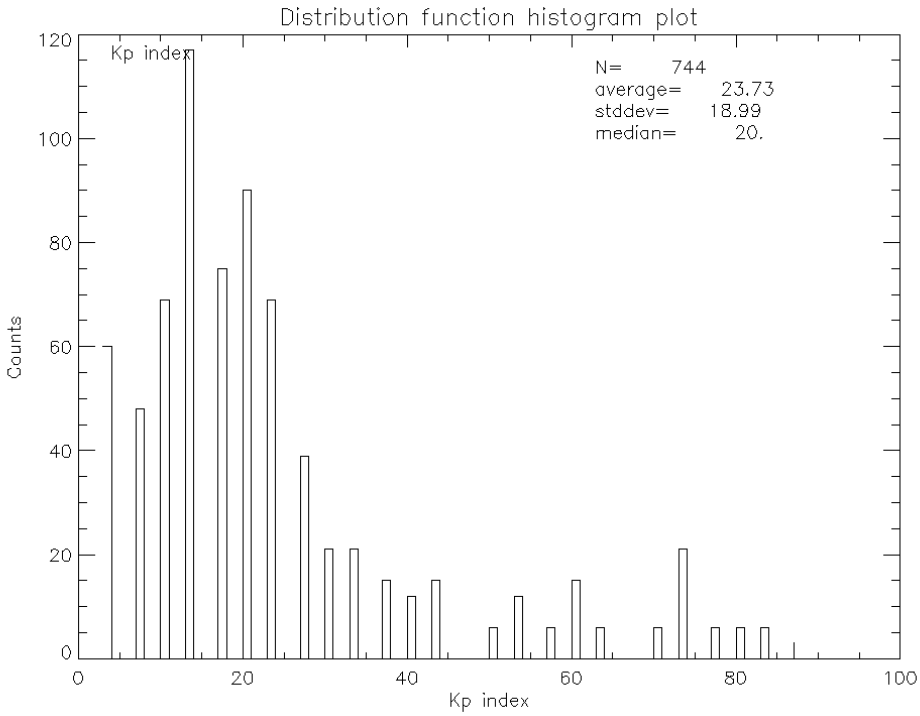


Figure 4: Distribution Function for July 2004

Here we find that the most common Kp is 11, so most common Kp is $Kp = 1.1$, and of there are no occurrences of Kp level 9,

Problem 7: Geomagnetic Storm

(i.)

We can recreate the Halloween Storm and obtain the following,



Figure 5: DST Index for the Halloween Storm

(ii.)

Using this information, we can calculate the ring current magnitude with, TODO: insert equations

$$B = \frac{\mu_0 I}{2R}, \; I = \frac{2RB}{\mu_0}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$$

$$B = 400 \times 10^{-9} \text{ T}$$

$$R = 5R_E = 5(6371) \times 10^3 \text{ m}$$

$$\boxed{\therefore I = 20 \times 10^6 \text{ A}}$$

Problem 8: Key Parameters in the Solar Wind Driving a Magnetic Storm

(i.)

The most critical parameter for describing solar wind is the interplanetary magnetic field. This is because for a geomagnetic storm to occur, the magnetic field field must point southward and also be sustained for an amount of time while also being stronger than usual, typically a result from coronal mass ejections. The interplanetary magnetic field essentially determines the conditions needed for a magnetic storm.

(ii.)

The positive surge that results from the magnetopause current of the Chapman Ferraro current. This is because when a CME hits Earth, the higher solar wind pressure compresses the magnetosphere inward, which is the magnetic storm's initial phase. The current forms because the solar wind gyrating around the magnetopause field lines. This cancels out the Earth's filed outside of the magnetosphere and increases the field inside of it, which leads to a eastward current as opposed to the usual westward current, causing a northward field and shows the surge in positive DST.

Problem 9: The Core

(i.)

Aaron Eckharts description of the Earth's core, electronics can be impacted by geomagnetic events, EM field does shield us from cosmic radiation.

(i.)

Microwave radiation will literally cook our planet, everybody on Earth is dead in a year, static discharge will not lead to superstorms.

Problem 10: Magnetospheric Multiscale